

Empirical Proof Record: REFUTED

2026-05-31T20:24:29.611071+00:00

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Proof ID: 436d337cb75c5f83b63d8f437a73f7b6...

Created: 2026-05-31T20:24:29.611071+00:00

What this means

This proof tests whether Kimera remembers the ORDER things happened, not just what happened — a filing cabinet (order-blind) versus a landscape you walk, where the path leaves a groove. It did not here: accumulation came out order-independent — the differentiator is permanence and path-depth, not order. (observed order_hysteresis = 0.0)

1. Claim

After the same documents in a different ORDER, the same identically-recognised query gets a different prime-address from Kimera (hysteresis) beyond its own run-noise, whereas a set-based retriever returns the identical ranking (order-blind by construction): $\text{order_hysteresis} = \text{mean}(\text{order_divergence}) - \text{mean}(\text{noise_floor}) > 0$.

- **Operationalisation:** Run 6 documents + 15 held-out probes through Kimera's entity target (batch) in order A, order A again (determinism control), and order B (shuffled). Per probe: $\text{order_divergence} = 1 - \text{Jaccard}(\text{prime_chain}|A, \text{prime_chain}|B)$; $\text{noise_floor} = 1 - \text{Jaccard}(\text{prime_chain}|A1, \text{prime_chain}|A2)$. $\text{order_hysteresis} = \text{mean}(\text{order_divergence}) - \text{mean}(\text{noise_floor})$. Baseline: TF-IDF retrieval order-divergence over the same docs/queries (= 0). **VALIDATED** iff $\text{order_hysteresis} > 0$ with a significant paired $\text{order_divergence} > \text{noise_floor}$ test.
- **Threshold:** $\text{order_hysteresis} > 0.0 \text{ jaccard_distance}$
- **H0:** H0: $\text{order_hysteresis} \leq 0$ — Kimera's response does not depend on document order beyond run-noise; accumulation is commutative, like a set-based retriever

- **H1:** H1: $\text{order_hysteresis} > 0$ — the same query after the same documents in a different order gets a different prime-address; Kimera carries the ORDER of experience (hysteresis) where RAG carries only the inventory

2. Verdict

REFUTED — observed 0.0

$\text{order_hysteresis} + 0.0000 = \text{Kimera order-divergence } 1.0000 - \text{noise-floor } 1.0000$ over 14 probes (prime-address); RAG (TF-IDF) order-divergence 0.0000 (set-based, order-blind by construction); manifold-state order-effect coupling $\Delta=2.374707$, mass $\Delta=0.982999$ (noise coupling $\Delta=3.411841$)

3. Evidence

Pillar	Statistic	Value	95% CI	Library
0.comparison.order_hysteresis	order_hysteresis	0.0	—	ophamin 0.115.2

4. Pre-registration

- Registered at: 2026-05-31T20:24:29.409326+00:00
- Config hash: 91269c9c9238ea1f32695ba2a90a0ad0...
- Data hash: b22b0bfc0b009088ddc2a6536a912cc7...

5. Signature

5c64f72023e5427a693cb15bcd525f57a3bf39b374a5bdaa231d8b68c86010ec